

Getting Permutations with Nested Loops: A Less-Than-Ideal Approach

Let's say we have a simple bicycle lock, as in [Figure 6-3](#), with a four-digit combination. The combination has 10,000 possible permutations of digits (0000 to 9999), but only one will unlock it. (They are called *combination locks*; however, in this context it'd be more accurate to call them *permutations with repetition locks*, since the order matters.)

Now let's say we have a much simpler lock with only the five letters A to E. We can calculate the number of possible combinations as 5^4 , or $5 \times 5 \times 5 \times 5$, or 625. A combination lock of k characters, each character selected from a set of n possibilities, is n^k . But getting a list of the combinations themselves is a bit more involved.